Q-Secure-P^2-SMA: Quantum-Secure Privacy-Preserving Smart Meter Authentication for Unbreakable Security in Smart Grid

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***Abstract*—Smart grids, steering an energy revolution through smart metering infrastructure, face cybersecurity threats and unauthorized access risks. Within these systems, classical authentication protocols validate identities, fostering trust and enabling access to authorized resources, while preventing unauthorized access. However, classical authentication protocols are predominantly dependent on public-key cryptography, rooted in computational hard problems such as integer factorization and discrete logs to uphold security. Nevertheless, these computational hard problems can be easily compromised by quantum adversaries. Furthermore, quantum-resistant smart meter authentication protocols are either deemed impractical or reliant on quantum hard problems, potentially lacking future-proof security. Moreover, maintaining SG entity privacy holds equal significance. Consequently, this article introduces the Quantum-Secure Privacy-Preserving Smart Meter Authentication (Q-Secure-P^2-SMA) protocol. It utilizes semi-quantum key distribution and a hash function to facilitate quantum-resistant metering data communication, with a focus on privacy and practicality. The security analysis and performance evaluation conducted for Q-Secure-P^2-SMA confirm its resilience against well-known security threats, showcasing improved efficiency compared to existing state-of-art-protocols across computation, communication, and energy overhead. Notably, the Q-Secure-P^2-SMA doesn't necessitate quantum capability among all SG entities, making it more practical and relevant in real-world scenarios.**

***Index Terms*—Authentication, privacy, smart meter, smart grid, semi-quantum key distribution, security threats, quantum threats.**

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I. Introduction

THE CONCEPT of the smart grid ushers in a transformative era of electrical infrastructure, fostering bidirectional communication and automation across traditional grid components [1]. Among the myriad applications of the smart grid, one particularly prominent use case lies in the realm of smart metering infrastructure (SMI). Smart metering infrastructure at its core, is composed of a dynamic interplay between a central control center (CC) and an array of smart meters (SMs) strategically placed within residential premises. This network orchestrates the collection of metering data, encompassing both electricity consumption and production figures [2]. The smart metering infrastructure gains depth through the integration of distributed energy resources at residential premises, such as solar panels, altering consumption with intermittent generation. This evolution lets households intermittently supply excess electricity to the grid, empowering consumers as active energy participants and bestowing grid operators and suppliers with invaluable insights. The meticulous aggregation of metering data provides suppliers with the capability to make accurate predictions about household electricity needs and energy production [3]. This predictive capability becomes crucial for optimizing electricity distribution, enabling precise resource allocation that minimizes supply losses and significantly enhances overall grid operational efficiency. Also, the smart grid enhances transparency in electricity distribution, allowing suppliers to streamline cost reconciliation with network and system operators, ushering in an era of accountability and precision in energy commerce.

Metering data serves as the anchor of all smart grid activities. The intrinsic economic importance attributed to electricity consumption data accentuates its susceptibility to various vulnerabilities and risks. Intermittently, the transmission of consumption data from smart meters is afflicted by intrusion and security attacks. This stems from the fact

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that this data not only holds immense value within the realm of energy management but also carries implications for broader socioeconomic and lifestyle insights. For example, an adversaries can exploit knowledge of electricity consumption profiles to identify opportune moments for intrusion (e.g., minimal consumption implying vacancy, thus facilitating potential break-ins). As such, the intricate interplay between the sensitivity of this data and its multifaceted utility creates a complex landscape wherein safeguarding its confidentiality and integrity while resisting numerous security attacks becomes paramount. The implementation of authentication mechanisms among smart grid entities serves to uphold data integrity, prevent unauthorized access, and maintain the security of the interconnected energy ecosystem. Still, privacy concerns stem from the intricate smart meter data, capturing detailed energy use, behavior, and occupancy. Aggregated analysis can reveal intimate routines, risking smart meter's owner privacy. Thus, considerations such as SM anonymity, SM message unlinkability, and SM identity privacy hold equal significance while securely exchanging metering data. Consequently, amidst the evolving landscape of data-driven utilities, addressing these privacy challenges is pivotal to fostering consumer trust and ensuring the sustainable evolution of the smart grid ecosystem. Therefore, the pivotal function of metering data in smart grid activities, coupled with its susceptibility to malicious intent and economic significance, underscores the need for robust security measures guaranteeing privacy-preserving metering data communication among authenticated SG entities.

A. Key Contributions

The primary contribution of this article is as follows:

1) First, we developed a mutual authentication and key exchange mechanism that leverages the benefits of semi-quantum key distribution to guarantee an unconditionally secure and trustworthy communication between SM and NAN gateway.
2) Q-Secure-P^2-SMA incorporates robust security features to protect against eavesdropping, man-in-the-middle attacks, replay attacks, and other potential threats. It guarantees message unlinkability, anonymity, identity-privacy, message integrity, non-repudiation and unconditional security. Additionally, we conducted a simulation study of Q-Secure-P^2-SMA using Qiskit.
3) Through an extensive performance evaluation, Q-Secure-P^2-SMA outperforms state-of-the-art solutions, demonstrating superior security and privacy attributes, including SM identity privacy and message unlinkability, while incurring lower computation and communication overhead. Additionally, Q-Secure-P^2-SMA exhibits energy efficiency, minimizing energy consumption during computational and communication tasks.
4) Furthermore, Q-Secure-P^2-SMA stands out as considerably more practically feasible in comparison to state-of-the-art schemes. This is achieved by not imposing the requirement for all entities to possess complete quantum capabilities. Specifically, in Q-Secure-P^2-SMA, we utilize semi-quantum key distribution, enabling the participation of a smart meter without quantum capabilities.

B. Organization

The following sections of this article are organized as follows: Section II delves into related research, while Section III provides an introduction to the system model and Section IV outlines the details of the proposed Q-Secure-P^2-SMA. Afterwards, Section V provides an in-depth examination of security aspects, and Section VI outlines performance assessment, including evaluation and simulation outcomes. Finally, Section VII provides concluding remarks.

II. Related Work

This section reviews authentication and key agreement schemes in the literature for improved smart grid security and privacy. A scheme for anonymous key distribution, employing identity-based encryption and identity-based signature techniques, has been developed for smart grid entities [4]. Additionally, Odelu et al. [5] discuss an efficient and provably secure authentication and key agreement approach tailored for smart grids, with consideration for the CK-adversary model. Furthermore, to facilitate privacy-based billing in smart grids, the studies by Gope and Sikdar [6], [7] present a privacy-preserving authentication and key agreement scheme, data aggregation scheme with dynamic demand response management. The scheme [8] aims to provide mutual authentication, anonymity, integrity, and session key exchange for smart grid entities employing hybrid cryptography. Nevertheless, many SG initiatives utilize public key cryptographic techniques, like elliptic curve cryptography (ECC), to enhance communication security [9], [10]. Later, the work [11] observes that the work presented in [10] lacks certain critical features such as protection against key compromise attacks, forward secrecy, anonymity and introduces a secure mutual authentication scheme for smart grids, which includes resistance against key compromise attacks, anonymity, and forward secrecy. These SG protocols employs traditional encryption and authentication techniques, relying on computational hard problems to guarantee SG communication security. While classical algorithms cannot solve hard problems like discrete log and integer factorization in polynomial time, quantum computers utilizing Shor's algorithm [12] can efficiently solve, posing a serious threat.

To address such threats, two approaches can be taken to design new quantum-resistant SG protocols. The new quantum-resistant SG protocols should either use quantum primitives that depends on laws of quantum mechanics or base their security on hard problems that remain secure against quantum computer threats [13]. Both techniques provide quantum resistance, but relying on lattice problems may not ensure future-proofing, as new quantum algorithms can emerge and pose threats, similar to computational hard problems. Several lattice-based schemes for SG communication security are detailed in [14], [15], [16], [17]. The work [18] uses

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NTRU based scheme for secure SG information exchange. However, these schemes does not guarantee future-proofing due to the potential emergence of new quantum algorithms. Furthermore, [19], [20], [21] describe schemes that utilize quantum primitives, leveraging the laws of quantum mechanics to ensure smart grid security. The work [19] uses quantum secret keys for smart grid authentication via quantum key distribution (QKD). These keys are integrated into the MQTT IoT protocol to secure distributed energy resource communications. Similarly, [20] introduces a cloud-based SG system model for advanced metering infrastructure (AMI) that integrates QKD to construct a quantum-secure cloud-based SG AMI. In [21], OTP is combined with quantum cryptography using a QKD terminal for secure key generation and distribution. The session key, produced by the quantum random number generator (QRNG) within the terminal, is distributed via a QKD protocol over a quantum channel, ensuring one-time use for each key. The work [22] uses BB84 QKD to guarantee quantum-secure communication among SG entities. While these SG schemes are designed to be future-proof, practical deployment necessitates significant updates due to their requirement for all SG entities to possess full quantum capabilities.

The existing literature on smart grid security often relies on hard problems, either computational or quantum, and frequently overlooks the need to safeguard the identity-privacy of SG entities. Additionally, the rapid advancement of quantum technology has reinforced existing quantum techniques, including those relying on the principles of quantum mechanics to ensure security. Nonetheless, many existing quantum-resistant smart grid security schemes, which rely on the principles of quantum mechanics to ensure security, demand full quantum capability from all entities, rendering it impractical and costly for upgrading the existing smart grid infrastructure. Consequently, this article introduces Q-Secure-P^2-SMA, a semi-quantum key distribution-based protocol that ensures privacy-preserving mutual authentication and communication for SG entities. Using semi-quantum key distribution compared to other quantum-resistant techniques offers several advantages, including: a) Enabling Q-Secure-P^2-SMA to provide resilience against both classical and quantum attacks without relying on computational or quantum hard problems. b) Making Q-Secure-P^2-SMA future-proof by harnessing the principles of quantum mechanics for security. c) Seamlessly integrating with the current SG infrastructure, eliminating the necessity for SM to possess complete quantum capabilities to participate in quantum-secured communication.

III. System Model

This section discusses the system model, adversary model, security and privacy requirements, and provides the adopted assumptions and background for the Q-Secure-P^2-SMA.

A. Smart Metering Infrastructure in Smart Grid

The smart grid's metering infrastructure encompasses SM, NAN gateway, and CC. In the proposed work, the CC assumes the duty of registering and authenticating both SMs and the NAN gateway. SMs are commonly situated in consumer locations like smart factories, homes, and buildings, effectively managing and transmitting energy consumption profiles. This consumption data is sent to the NAN gateway for periodic electricity billing in line with consumer pricing strategies. Significantly, the SM supports bi-directional communication, capable of receiving demand-response signals from the NAN gateway. It also interfaces with in-house displays, offering real-time appliance electricity consumption statistics to consumers. While the NAN gateway and CC possess quantum capabilities, the SMs remain classical. The NAN gateway aggregates consumption readings from various SMs and ensures secure data transmission to the CC. Acting as an intermediary, it facilitates communication between the upper layer (CC) and lower layer (smart meter). Q-Secure-P^2-SMA approach aligns naturally with the existing smart metering infrastructure. The utilization of a classical smart meter removes the need for a quantum channel in our quantum-secure communication design. Consequently, the transition to quantum security becomes a natural enhancement that complements the current smart metering infrastructure. This approach not only streamlines the implementation process but also enhances the feasibility of enabling quantum-secure communication between SM and NAN gateway into the established infrastructure. This deliberate decision empowers the existing infrastructure to concentrate efforts on upgrading the NAN gateway and CC from classical to quantum capabilities. This transition can be seamlessly accomplished by connecting the NAN gateway and CC through a fiber optics channel. Fig. 1 illustrates the system model of Q-Secure-P^2-SMA. We assume that both the NAN gateway and CC, possessing quantum capabilities, establish a secure data transfer connection using semi-quantum key distribution methods. However, it's worth noting that quantum key distribution techniques can also serve this purpose.

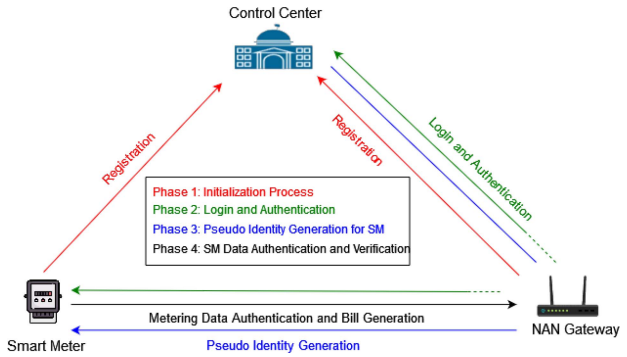

Fig. 1. System Model of Q-Secure-P^2-SMA.

B. Adversary Model

The adversary model applied in Q-Secure-P^2-SMA incorporates secured interactions among SMs, NAN gateway, and CC during the SM registration process. Similarly, the registration of NAN gateway with CC occurs within a secure channel. However, Q-Secure-P^2-SMA's protocol execution unfolds over an insecure public channel. The protocol adheres to the Dolev-Yao model [23], where an adversary can intercept

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TABLE I
SECURITY AND PRIVACY REQUIREMENTS

Notations	Definition
Mutual authentication	Verification of authenticity can be mutually performed for both the SM and CC.
Session key agreement	A shared session key is generated by the SM and NAN gateway to ensure secure data exchange.
Resistance to attacks	The scheme is constructed to resist various attack types, encompassing replay, Man-in-the-Middle (MITM), and impersonation attacks.
Anonymity	The identities of both the SM and NAN gateway must remain undisclosed to potential attackers.
Forward secrecy	Q-Secure-P^2-SMA provides assurance that even if an attacker acquires the secret keys for a given session, they cannot access the secret keys of preceding sessions.
No clock synchronization	Q-Secure-P^2-SMA does not address issues that arises from time delays and clock synchronization.
SM identity privacy	Q-Secure-P^2-SMA safeguards the privacy of the actual identity of the SM.
SM message unlinkability	Q-Secure-P^2-SMA ensures the absence of linkage between messages originating from the same SM.
SM traceability	Q-Secure-P^2-SMA facilitates the ability to trace the SM in case of misreporting or misconduct.
Unconditional security	Q-Secure-P^2-SMA attains security without relying on computational or quantum hard problems.

messages exchanged between SMs, NAN gateway, and/or CC, exploiting the public channel for various attacks like forgery, replay, and man-in-the-middle. Adversaries may exploit the public channel to acquire real identity information of SMs and impersonate SMs for unauthorized benefits, necessitating stringent identity privacy for SMs. Malicious intent can lead adversaries to extract identity details of individual SMs during message's exchange via the insecure channel among SMs, NAN Gateway, and/or CC. Moreover, the adversary might attempt to discern whether two messages originate from the same SM. Furthermore, malicious SMs might provide fake metering data to incur lower charging fees.

C. Security and Privacy Requirements

With reference to the evaluation guidelines [24] for quantum-resistant protocols, we formulate Q-Secure-P^2-SMA to fulfill the security and privacy prerequisites outlined in Table I. The privacy criteria detailed in Table I, encompassing aspects like SM message unlinkability, SM identity privacy, and anonymity, are relevant to evaluating conditional privacy-preserving characteristics. Meanwhile, the remaining requisites primarily concern the scrutiny of security features.

D. Assumptions

This subsection outlines the underlying assumptions that constitute the foundation of Q-Secure-P^2-SMA, as follows:

1) The CC is established as the trusted entity, possessing substantial computation and memory capabilities, while the NAN gateway is categorized as a semi-trusted entity, exhibiting honesty but curiosity. The SMs are considered as resource-constrained entities with restricted computation and memory capabilities [9].
2) The SMs are equipped with devices to capture biological attributes like retina, facial, or fingerprint details of their registered owner [22].
3) The entities within the smart metering infrastructure network, including the NAN gateway and CC, are equipped with quantum communication capabilities, whereas SMs lack such capabilities. Additionally, the clocks of the SMs, NAN gateway, and CC are synchronized [10].
4) The SMs deployed within the considered smart metering infrastructure encompass both smart electricity meters and smart gas meters. This alignment follows the practices of numerous existing SMIs, as observed in countries like the Netherlands and Great Britain, where both types of SMs are still in use [8]. Significantly, for smart electricity meters, the reduction in energy consumption is paramount due to their continuous operation. This holds particular relevance as the projected deployment of over 1.1 billion units in the Asia-Pacific region by 2027 [25] could potentially influence electricity consumption trends in the region.

E. Background of Proposed Work

The proposed Q-Secure-P^2-SMA approach relies on semi-quantum key distribution, a method used to distribute a key between classical entities (classical SM) and quantum entities (like quantum NAN gateway and CC within the smart grid ecosystem). The classical entity SM is capable of: (1) Measurement of qubits in Z basis (SIFT); (2) Reflecting the qubits without any alteration (CTRL); (3) Reordering the qubits. We assume the parameters used for the semi-quantum key distribution protocol are: $n = 2$ and $\delta = \frac{1}{8}$. First, quantum CC or NAN gateway is responsible for the generation of a random state $|\psi_0\rangle$ of $N = \lceil(8n(1+\delta)\rceil$. CC sends this random state to classical SM. Afterwards, SM randomly performs SIFT or CTRL operation on the qubits. The CTRL qubits are sent back to CC in random order. CC then measures the CTRL qubits and verifies them against their initial values in $|\psi_0\rangle$. The verification of CTRL bits must yield an error rate below a specified threshold denoted as P_{CTRL}, ensuring the absence of an eavesdropper during the communication from SM to CC. Now, in the absence of an eavesdropper and any noise, CC should measure the CTRL qubit state $|0\rangle, |1\rangle, |+\rangle, |-\rangle, |1\rangle, |-\rangle, |0\rangle, |1\rangle, |-\rangle$ for successful verification. Following this, the protocol moves to the SIFT qubits encoded in the Z basis and designates them as $|\psi_1\rangle$. Afterwards, SM picks n random bits, which are termed TEST bits, from the set $|\psi_1\rangle$. CC then discloses the initial state of these bits, and SM verifies whether the error rate is below a threshold of P_{TEST} and aggregates them into a string denoted as v. This verification ensures that there was no eavesdropping during the transmission of these qubits from CC to SM. After ensuring no eavesdropper in the entire process of communication, CC and SM pick the first n bits from string v called as INFO bits. INFO bits are used as random seed for generation of the $SharedKey_X$. The generation of INFO bits is illustrated in Table II. Besides, Q-Secure-P^2-SMA encodes I_{SM_i} into $I_{Enc_{SM_i}}$ by performing an XOR operation between I_{SM_i} and the SM owner's biological data, such as biometric or facial information.

IV. PROPOSED WORK

The proposed Q-Secure-P^2-SMA scheme capitalizes on the advantages offered by semi-quantum key distribution (SQKD)

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TABLE II
GENERATION OF SHARED KEY

Qubit Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Message ($\|\psi_0\rangle$)	$\|+\rangle$	$\|0\rangle$	$\|0\rangle$	$\|+\rangle$	$\|0\rangle$	$\|1\rangle$	$\|-\rangle$	$\|1\rangle$	$\|+\rangle$	$\|-\rangle$	$\|-\rangle$	$\|1\rangle$	$\|+\rangle$	$\|-\rangle$	$\|0\rangle$	$\|0\rangle$	$\|1\rangle$	$\|-\rangle$
CTRL Bits					$\|0\rangle$			$\|1\rangle$	$\|+\rangle$		$\|-\rangle$	$\|1\rangle$		$\|-\rangle$	$\|0\rangle$		$\|1\rangle$	$\|-\rangle$
SIFT Bits	$\|+\rangle$	$\|0\rangle$	$\|0\rangle$	$\|+\rangle$		$\|1\rangle$	$\|-\rangle$			$\|-\rangle$			$\|+\rangle$			$\|0\rangle$		
$\|\psi_1\rangle$		$\|0\rangle$	$\|0\rangle$			$\|1\rangle$										$\|0\rangle$		
TEST Bits						$\|1\rangle$										$\|0\rangle$		
v		0	0															
INFO Bits		0	0															

TABLE III
NOTATIONS USED IN Q-SECURE-P^2-SMA

Notations	Descriptions
I_{SM_i}	Representing identity of ith SM
$I_{Enc_{SM_i}}$	Representing encrypted identity of ith SM
I_{NAN_j}	Representing identity of jth NAN gateway
I_X	Used to reference I_{SM_i} or I_{NAN_j}
SM_i	Representing i^{th} SM
NAN_j	Representing j^{th} NAN gateway
X	Used to reference SM_i or NAN_j
M_{SM_i}	Randomly chosen measurement sequence corresponding to i^{th} SM
T_{SM}, T_{NAN}	Time-stamps
$SharedKey_X$	Used to reference $SharedKey_{SM_i}$ or $SharedKey_{NAN_j}$
$SharedKey_{SM_i}$	Secret key of ith SM
$SharedKey_{NAN_j}$	Secret key of jth NAN gateway
P_{CTRL}	Error acceptable in validation of CTRL bits of SQKD
P_{TEST}	Error acceptable in validation of TEST bits of SQKD

and classical identity authentication to strengthened the SMI security. Initially, both the SM and NAN gateway initiate registration with the control center to acquire their respective secret keys. Subsequently, the SM undergoes authentication with the CC via the NAN gateway employing the concept of mutual identity authentication scheme detailed in [26], resulting in the issuance of a pseudo identity for SM. Corresponding to this issued pseudo identity, the SM and NAN gateway generate a session key, which, in conjunction with the pseudo identity, enables secure and privacy-preserving metering data communication. The proposed Q-Secure-P^2-SMA is organized into four phases: a) Initialization Process, b) Login and Authentication, c) Pseudo-identity Generation for SM, and d) SM Data Authentication and Verification. The notations and their corresponding explanations for Q-Secure-P^2-SMA are available in Table III.

A. Phase 1: Initialization Process

This phase involves initializing the entities within the smart metering infrastructure, such as the smart meter, NAN gateway, and control center, before their deployment. The specific registration process unfolds as follows:

1) Control Center Initialization: First, CC undergoes initialization. Subsequently, CC initiates the registration process for both the smart meter and NAN gateway using SQKD. After a successful registration process, the CC proceeds to update the databases of both the NAN gateway and SMs with their respective secret keys (each of length *l*) in the following manner:

- Initially, CC selects an even natural number *n* as the preferred INFO string length, along with a predetermined parameter δ.
- CC transmits $N = \lceil (8n(1+\delta) \rceil$ qubits. For each qubit, they make a random choice between sending it in the Hadamard basis (H) or the computational basis (Z) and for each chosen basis, they transmit random bits.
- Upon the arrival of each qubit, entity *X* makes a random decision between measuring it (SIFT) or reflecting it (CTRL). Furthermore, *X* reorders the reflected qubits randomly, ensuring that neither the CC nor any potential eavesdropper can discern which qubits were reflected.
- CC gathers the reflected qubits in a quantum memory and publicly discloses the Z bits used. Entity *X* then discloses the CTRL qubits and the specific sequence in which CTRL qubits were reflected. Finally, all the returned CTRL qubits are measured by the CC in their originally prepared basis.
- CC assesses the error rate on the CTRL bits. If either the Z error-rate or the X error-rate surpasses a predetermined threshold denoted as P_{CTRL}, the protocol is terminated.
- CC randomly selects *n* SIFT bits to serve as TEST bits. The chosen bits are then publicly disclosed. X reveals the values of these TEST bits. CC then proceeds to evaluate the error rate of the TEST bits, and in case it surpasses a predefined threshold denoted as P_{TEST}, the protocol is terminated. Alternatively, if the error rate falls within the threshold, the remaining SIFT bits are aggregated into a string denoted as *v*.
- CC and X select the initial *n* bits within the *v* string as the INFO bits. In the absence of errors or eavesdropping, CC and X possess an identical INFO string. This INFO string serves as the seed for generating the secret key of length *l* bits.
- Otherwise, CC publishes the privacy amplification (PA) data and error correction code. Both CC and X derive the *l*-bit final key from the INFO string using the published information. Consequently, CC and X possess an identical shared key $SharedKey_X$.

2) NAN Gateway Registration: The NAN gateway initiates its registration with the control center before assuming the responsibility of managing smart meters within a designated region. During this registration phase, the NAN Gateway is assigned a distinctive identity denoted as I_{NAN_j} by the control center. Subsequently, the NAN Gateway proceeds with the prescribed registration steps, culminating in the acquisition of the shared secret key $SharedKey_{NAN_j}$. Consequently, the NAN Gateway parameters following a successful registration encompass: I_{NAN_j}, $SharedKey_{NAN_j}$.

3) Smart Meter Registration: The SM must complete a registration process with the control center before offering household electricity management services. In this procedure, the SM obtains a distinct identity, I_{SM_i}, from the CC.

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Following that, SM_i employs biological data such as biometric or facial information of its registered owner to encode I_{SM_i} into $I_{Enc_{SM_i}}$. Subsequently, SM_i proceeds with the registration procedure as outlined, ultimately obtaining the shared secret key $SharedKey_{SM_i}$. After successful registration, the SM parameters includes: I_{SM_i}, $I_{Enc_{SM_i}}$, $SharedKey_{SM_i}$.

B. Phase 2: Login and Authentication

The Q-Secure-P^2-SMA scheme facilitates the authentication of both smart meters and NAN gateways by the control center prior to engaging in bi-directional communication. The procedure involves initial verification of the NAN gateway's authenticity, followed by a subsequent SM legitimacy assessment through the verified NAN gateway. In accordance with the proposed Q-Secure-P^2-SMA scheme's system model, CC possesses robust quantum communication capabilities, while X is confined to classical capabilities, X's actions are constrained within four distinct operations when engaging with a segment of the quantum channel: (a) measurement – quantum states are measured in the Z basis $(|0\rangle, |1\rangle)$; (b) preparation–fresh qubits are prepared in the Z basis; (c) reflection–qubits are reflected to the quantum CC without disruption; (d) reordering–qubits are rearranged utilizing distinct delay lines. Moreover, during the registration process, CC and X pre-exchange the classical secret key $(SharedKey_X)$ sequence $K(K_{2i-1}K_{2i} \in 00, 01, 10, 11, i = 1, 2, \ldots, n)$ using the SQKD protocol. Assuming a noiseless and lossless quantum channel, the process of mutual authentication between CC and X is outlined as follows:

- CC prepares a sequence of n Bell states, randomly assigning each state to one of the four Bell states: $|\varphi^+\rangle_{12}, |\varphi^-\rangle_{12}, |\phi^+\rangle_{12}, |\phi^-\rangle_{12}$, where $\left|\phi^{+/-}\right\rangle_{ij}$ represents i^{th} and j^{th} qubit respectively. CC then divides these Bell states into two sequences: the first qubits constitute the home sequence S_H $(S_H = S_{H1}, S_{H2}, \ldots, S_{Hn})$, and the second qubits form the traveling sequence S_T $(S_T = S_{T1}, S_{T2}, \ldots, S_{Tn})$. Additionally, CC generates $\frac{n}{2}$ decoy qubits (D) based on the pre-shared key K. The rules for generation are outlined as follows: For i = 1, 2, . . . , 2n:
$$D_i = \begin{cases} |0\rangle \text{ or } |1\rangle, & K_{2i-1}K_{2i} \in \{00, 01\} \\ |+\rangle \text{ or } |-\rangle, & K_{2i-1}K_{2i} \in \{10, 11\} \end{cases}$$
CC inserts the decoy qubits into S_T randomly, resulting in the sequence S_{TD}. CC maintains a record of the inserted positions and the initial states of these decoy qubits. Subsequently, it sends the sequence S_{TD} to X.
- Upon receiving the sequence S_{TD}, X promptly acknowledges receipt to CC. Subsequently, CC reveals the positions where the decoy qubits were inserted. CC and X then execute the designated operations on the decoy qubits as per the pre-shared key sequence K. If $K_{2i-1}K_{2i} \in \{00, 01\}$, X measures the decoy qubits D_i in the Z basis and publicly shares the measurement outcomes. Conversely, if $K_{2i-1}K_{2i} \in \{10, 11\}$, X reshuffles the decoy qubits D_i and transmits them back to CC. Upon receiving the retransmitted decoy qubits, CC inquires about the original order, reinstates it, and subsequently measures them in the X basis.
- CC cross-references all measurement outcomes with the initial states of the decoy qubits. If any inconsistencies arise between the measurement outcomes and the initial states, the protocol is reset from the initial phase. If, however, the measurement outcomes align with the initial states, X gains access to sequence S_T and proceeds to the subsequent phase.
- Now, CC and X perform Z basis measurements on qubits at S_H and S_T, converting the results into classical outcomes R_A and R_B.
- CC announces the initial states of the Bell states in preparation. Leveraging the Z basis measurement outcomes and the initially announced Bell states by CC, X can infer CC's Z basis measurement outcomes as R_A^*. Similarly, CC can deduce X's Z basis measurement outcomes as R_B^*. Now, X conducts bitwise XOR operations on R_A^* and K, as well as R_B and K, resulting in I_A^* and I_B. Likewise, CC derives I_A and I_B^*. The detailed steps can be summarized as follows: For i = 1, 2, . . . , n:
$$\begin{aligned} R_{Ai} \oplus K_i &= I_{Ai}, \quad I_{Ai} = \{I_{A1}, I_{A2}, \ldots, I_{An}\} \\ R_{Bi} \oplus K_i &= I_{Bi}, \quad I_{Bi} = \{I_{B1}, I_{B2}, \ldots, I_{Bn}\} \\ R_{Ai}^* \oplus K_i &= I_{Ai}^*, \quad I_{Ai}^* = \{I_{A1}^*, I_{A2}^*, \ldots, I_{An}^*\} \\ R_{Bi}^* \oplus K_i &= I_{Bi}^*, \quad I_{Bi}^* = \{I_{B1}^*, I_{B2}^*, \ldots, I_{Bn}^*\} \end{aligned}$$
- Subsequently, X shares I_A^* with CC, and CC shares I_B^* with X. Finally, if $I_A = I_A^*$, CC successfully verifies the authenticity of X, confirming X as a legitimate identity. Conversely, if $I_B = I_B^*$, X successfully authenticates CC; otherwise, CC's authenticity is not established and connection is discarded.

Initially, the SM_i transmits its preauthentication request $PreAuthReq_{SM_i} = HMAC(SharedKey_{SM_i}, I_{Enc_{SM_i}})||I_{Enc_{SM_i}}||TS$ to validate its identity through NAN_j. The NAN gateway forwards the $PreAuthReq_{NAN_j} = PreAuthReq_{SM_i}||HMAC(SharedKey_{NAN_j}, I_{NAN_j})||I_{NAN_j}||TS$ to CC, which subsequently undergoes the authentication process outlined earlier. Before confirming the identity of SM_i, the control center verifies the NAN gateway identity; I_{NAN_j}. Upon successfully authenticating NAN_j, CC commences the authentication process for SM_i following the aforementioned authentication procedure. Once SM_i authentication is successful, CC proceeds to initiate the generation of a pseudo-identity for the smart meter.

C. Phase 3: Pseudo Identity Generation for SM

CC generates a random number R and produces pseudo identity $PSID_{SM_i} = E_{SharedKey_{SM_i}}[I_{Enc_{SM_i}}||R]$ after successfully authenticating SM_i. Afterward, CC transfers $ACKPreAuthReq_{NAN_j}$ back to NAN_j. Next, NAN_j forwards the $ACKPreAuthReq_{SM_i}$ to SM_i and initiates the session key generation process for SM_i using the corresponding pseudo identity $PSID_{SM_i}$. The following steps outline the process where:

$$AckPreAuthReq_{SM_i} = PSID_{SM_i}||TS$$

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- Initially, upon receiving $AckPreAuthReq_{SM_i}$ from NAN_j, SM_i extracts $PSID_{SM_i}$ and proceeds to verify the authenticity of $PSID_{SM_i}$ by conducting decryption using $SharedKey_{SM_i}$. After successful $PSID_{SM_i}$ authenticity verification, NAN_j selects an even natural number *n* as the preferred INFO string length, along with a predetermined parameter δ.
- Afterward, NAN_j transmits $N = \lceil (8n(1+\delta) \rceil$ qubits. For each qubit, they make a random choice between sending it in the Hadamard basis (H) or the computational basis (Z), and for each chosen basis, they transmit random bits.
- Upon the arrival of each qubit, SM_i makes a random decision between measuring it (SIFT) or reflecting it (CTRL). Furthermore, SM_i reorders the reflected qubits randomly, ensuring that neither the NAN gateway nor any eavesdropper can discern which qubits were reflected.
- NAN_j gathers the reflected qubits in a quantum memory and publicly discloses the Z bits used. Entity SM_i then discloses the CTRL qubits and the specific sequence in which CTRL qubits were reflected. Finally, all the returned CTRL qubits are measured by the NAN_j in their originally prepared basis.
- Subsequently, NAN_j assesses the error rate on the CTRL bits. If either the Z error-rate or X error-rate surpasses a predetermined threshold denoted as P_{CTRL}, the protocol is terminated.
- NAN_j randomly selects *n* SIFT bits to be designated as TEST bits. These selected bits are made public, and SM_i discloses the values of these TEST bits. NAN_j then proceeds to evaluate the error rate of the TEST bits, and in case it surpasses a predefined threshold denoted as P_{TEST}, the protocol is terminated. Alternatively, if the error rate is within the threshold, the remaining SIFT bits are aggregated into a string denoted as *v*.
- NAN_j and SM_i select the initial *n* bits within the *v* string as the INFO bits. In the absence of errors or eavesdropping, NAN_j and SM_i possess an identical INFO string. This INFO string serves as the seed for generating the secret key of length *l* bits.
- Otherwise, NAN_j publishes the privacy amplification (PA) data and error correction code. Both NAN_j and SM_i derive the *l*-bit final key from the INFO string using the published information. Consequently, NAN_j and SM_i possess an identical session key $Sessionkey_{SM_i}$.

The successful execution of the last two phases results in SM_i authenticating itself, obtaining the pseudo identity $PSID_{SM_i}$ from the control center, and establishing a session key with NAN_j based on $PSID_{SM_i}$, facilitating further privacy-preserving SM communication.

D. Phase 4: SM Data Authentication and Verification

First, SM_i transmits a tuple $BillReq_{SM_i} = (MSG, MD_{Current}, MCU, PSID_{SM_i}, T_{SM})$ to NAN_j for the purpose of reporting current metering data $(MD_{Current})$ and receiving the monthly consumption bill $(Bill_{SM_i})$. The *MSG* within the tuple $BillReq_{SM_i}$ is an HMAC operation that involves the monthly consumption usage (MCU) along with the session key $Sessionkey_{SM_i}$. The MCU is a composition of metering data $(MD_{Current})$, $PSID_{SM_i}$, and the timestamp T_{SM}, which serves to protect the actual identity I_{SM_i} and prevent replay attacks.

$$MSG = HMAC\big(Sessionkey_{SM_i}, MCU\big)$$
$$MCU = MD_{Current} || PSID_{SM_i} || T_{SM}$$

Lastly, NAN_j validates the received message (MSG) using the previously shared session key $Sessionkey_{SM_i}$ between NAN_j and SM_i during session key generation. Specifically, NAN_j extracts both $Sessionkey_{SM_i}$ and $PSID_{SM_i}$ from MSG and cross-references $Sessionkey_{SM_i}$ in its database corresponding to $PSID_{SM_i}$. If the search yields a positive result, the MSG is accepted as authentic; otherwise, it is discarded. In cases of authentic MSG, NAN_j produces and transmits the monthly consumption bill $(Bill_{SM_i})$ to SM_i according to the consumption pricing policy and MD_{Nett}. Afterwards, NAN_j updates $MD_{Current}$ as $MD_{Previous}$ in its database corresponding to $PSID_{SM_i}$.

$$MD_{Nett} = MD_{Current} - MD_{Previous}$$

V. Security Evaluation

This section assess the security features of Q-Secure-P^2-SMA, as indicated in Table IV, compared to prior research [4], [5], [6], [8], [10], [11], [18], [22]. Provable security is a methodology that relies on established complexity-theoretic assumptions, such as the computational discrete log problem's intractability. It employs formal techniques to achieve specific security objectives while assessing and demonstrating the security of cryptographic protocols [24]. However, it is essential to note that while provable security is a powerful tool, it is not immune to limitations. Security proofs relying on formal methods may sometimes lack complete or precise security models or exhibit inaccuracies in the reduction of security proofs. These limitations can result in security proofs that do not offer comprehensive protection against practical threats. Furthermore, the security proof correctness can significantly rely on the expertise of the individual providing the proof [27]. Even when employing provable security techniques with accurate security models and correct reductions, there is no absolute guarantee of flawless security analysis. As a result, the traditional heuristic approach continues to play a significant role in demonstrating the versatility and security of newly designed cryptographic protocols. In line with this understanding, this section assesses a heuristic approach for establishing the security of the proposed Q-Secure-P^2-SMA scheme designed for smart metering infrastructure. This assessment conforms to the threat model and evaluation criteria specified in Table I.

A. Security Features Supported

Q-Secure-P^2-SMA uses semi-quantum key distribution and reports the same theoretical security as described in [28]. Nevertheless, it's essential to consider potential threats from both external and internal attackers who could disrupt SG communication by launching attacks or malicious code. To

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address these concerns, the proposed work aims to fulfill the following security objectives:

Proposition 1: Q-Secure-P^2-SMA features Resistance to Measure and Resend Attack

Proof: Let the two-dimensional Hilbert space modeling the qubit is represented by $\mathrm{H_T}$, and adversary's (A) ancilla for one iteration of the Q-Secure-P^2-SMA is denoted by $\mathrm{H_A}$. Consider a pair of unitary attack operators, denoted as ($\mathrm{U_E}$, $\mathrm{U_F}$), both applied on ($\mathrm{H_T} \otimes \mathrm{H_A}$). $\mathrm{U_E}$ is employed for the attack in the direction of NAN_j to SM_i, while $\mathrm{U_F}$ is used for attacking the reverse direction, SM_i to NAN_j. This characterizes the standard collective attack, where A employs the identical unitary operators, $\mathrm{U_E}$ and $\mathrm{U_F}$, in each iteration of the quantum communication stage. In each of these iterations, A employs a distinct 'copy' of $\mathrm{U_E}$. This assumes that the iteration contributes to the raw key generation, specifically assuming that NAN_j sent X basis or Z basis states, SM_i measured and resent the qubit, and NAN_j measured it in the X basis or Z basis. To compute the key rate for Q-Secure-P^2-SMA, it is necessary to build the density operator for a single iteration. In Q-Secure-P^2-SMA, we consider both Z basis and X basis noises. However, we do not account for Y basis noises or the identity operator, as they involve both phase and bit flip aspects together, leading to increased complexity in its estimation [29]. ■

- *Using Z-basis Noise:*

1) Given this scenario, NAN_j prepares a qubit in the form of $|0\rangle$ or $|1\rangle$, with each having a probability of $\frac{1}{2}$. A receives a mixed state ρ_1 in $\mathrm{H_T}$,

$$\rho_1 = 1/2|0\rangle\langle 0|_\mathrm{T} + 1/2|1\rangle\langle 1|_\mathrm{T} \tag{1}$$

2) It is assumed that A's ancilla is initialized to a "zero" state in $\mathrm{H_A}$ space, and then attacked using $\mathrm{U_E}$, an operator that acts on basis states. The mixed state is obtained as ρ_2,

$$\rho_2 = \mathrm{U_E}\big(1/2|0\rangle\langle 0|_\mathrm{T} + 1/2|1\rangle\langle 1|_\mathrm{T}\big)\, U_\mathrm{E}^\dagger \tag{2}$$

where, $\mathrm{U_E}|00\rangle_\mathrm{TA} = |0\mathrm{e}_0\rangle_\mathrm{TA} + |1\mathrm{e}_1\rangle_\mathrm{TA}$ and $\mathrm{U_E}|10\rangle_\mathrm{TA} = |0\mathrm{e}_2\rangle_\mathrm{TA} + |1\mathrm{e}_3\rangle_\mathrm{TA}$ and all $|\mathrm{e_i}\rangle$ are the arbitrary states in the adversary's hilbert space $\mathrm{H_A}$, with applied conditions, $\langle \mathrm{e}_0|\mathrm{e}_0\rangle + \langle \mathrm{e}_1|\mathrm{e}_1\rangle = 1$; $\langle \mathrm{e}_2|\mathrm{e}_2\rangle + \langle \mathrm{e}_3|\mathrm{e}_3\rangle = 1$; $\langle \mathrm{e}_0|\mathrm{e}_1\rangle + \langle \mathrm{e}_2|\mathrm{e}_3\rangle = 0$

3) After this operation, A passes the qubit to SM_i who can perform a Z or X basis measurement, known as SHIFT, recording the outcome as its raw key bit, and then resend this outcome to NAN_j. Alternatively, SM_i can apply CTRL, which involves resending the qubit to NAN_j without measuring it or making any changes. Presently, the system is in the mixed state:

SHIFT:

$$\rho_3^\mathrm{S} = \frac{1}{2}|0\rangle\langle 0|SM_\mathrm{i} \otimes \big(|0e_2\rangle\langle 0e_2|_\mathrm{TA} + |0e_0\rangle\langle 0e_0|_\mathrm{TA}\big) + \frac{1}{2}|1\rangle\langle 1|_{SM_\mathrm{i}} \otimes \big(|1e_1\rangle\langle 1e_1|_\mathrm{TA} + |1e_3\rangle\langle 1e_3|_\mathrm{TA}\big) \tag{3}$$

CTRL:

$$\rho_3^\mathrm{C} = \rho_2 \tag{4}$$

4) A acquires the transit qubit upon its return and employs the second attack operator, $\mathrm{U_E}$. The action of this operator on states can be described as $|i, e_\mathrm{j}\rangle$ which follows:

$$U_\mathrm{F}|i, e_\mathrm{j}\rangle_\mathrm{TA} = |0, e^0_\mathrm{i,j}\rangle_\mathrm{TA} + |1, e^1_\mathrm{i,j}\rangle_\mathrm{TA} \tag{5}$$

with applied conditions as, $\langle e_\mathrm{j}|e_\mathrm{j}\rangle = \langle e^0_\mathrm{i,j}|e^0_\mathrm{i,j}\rangle + \langle e^1_\mathrm{i,j}|e^1_\mathrm{i,j}\rangle$; where, i,j = 0,1

SHIFT:

$$\begin{aligned}\rho_4^\mathrm{S} = \frac{1}{2}|0\rangle\langle 0|_{SM_\mathrm{i}} \otimes \Big(|e^0_{0,0}\rangle\langle e^0_{0,0}|_\mathrm{TA} + \Big(|e^1_{0,0}\rangle\langle e^1_{0,0}|_\mathrm{TA} \\ + \Big(|e^0_{0,2}\rangle\langle e^0_{0,2}|_\mathrm{TA} + \Big(|e^1_{0,2}\rangle\langle e^1_{0,2}|_\mathrm{TA}\Big) + \frac{1}{2}|1\rangle \\ \langle 1|_{SM_\mathrm{i}} \otimes \Big(|e^0_{1,1}\rangle\langle e^0_{1,1}|_\mathrm{TA} + \Big(|e^1_{1,1}\rangle\langle e^1_{1,1}|_\mathrm{TA} \\ + \Big(|e^0_{1,3}\rangle\langle e^0_{1,3}|_\mathrm{TA} + \Big(|e^1_{1,3}\rangle\langle e^1_{1,3}|_\mathrm{TA}\Big)\end{aligned} \tag{6}$$

CTRL:

$$\rho_4^\mathrm{C} = U_\mathrm{F}\big(|e, 0\rangle\langle e, 0|\big)\, U_\mathrm{F}{}^\dagger \tag{7}$$

5) Now, SM_i measures in either the Z or X basis at random, leading to final mixed-state density operator as:

$$\begin{aligned}\rho_5^{\mathrm{S-Z}} = |0\rangle\langle 0|_{NAN_\mathrm{j}} \otimes \frac{1}{2}|0\rangle\langle 0|_{SM_\mathrm{i}} \otimes \Big(|e^0_{0,0}\rangle\langle e^0_{0,0}|_\mathrm{TA} \\ + |e^1_{0,0}\rangle\langle e^1_{0,0}|_\mathrm{TA} + |e^0_{0,2}\rangle\langle e^0_{0,2}|_\mathrm{TA} \\ + |e^1_{0,2}\rangle\langle e^1_{0,2}|_\mathrm{TA}\Big) + |1\rangle\langle 1|_{NAN_\mathrm{j}} \otimes \frac{1}{2}|1\rangle \\ \langle 1|_{SM_\mathrm{i}} \otimes \Big(|e^0_{1,1}\rangle\langle e^0_{1,1}|_\mathrm{TA} + |e^1_{1,1}\rangle\langle e^1_{1,1}|_\mathrm{TA} \\ + |e^0_{1,3}\rangle\langle e^0_{1,3}|_\mathrm{TA} + |e^1_{1,3}\rangle\langle e^1_{1,3}|_\mathrm{TA}\Big)\end{aligned} \tag{8}$$

Notably, SM_i and NAN_j estimate the Z basis noise in both the transmitting and receiving channels as part of the parameter estimation stage. Specifically, they determine the value of $p_\mathrm{i,j,k}$. This probability represents the likelihood that when NAN_j initially sends $|i\rangle$, then SM_i measures $|j\rangle$, and NAN_j measures $|k\rangle$. For instance, in the absence of noise in the Z basis, it should be the case that $p_{0,0,0} = 1$. Also, these parameters can be utilized for estimating the value of $\langle e_\mathrm{j}|e_\mathrm{j}\rangle$.

$$\begin{aligned}&p_{0,1,0} \\ &= Prob\big(SM_\mathrm{i} measures|1\rangle\big)\, and Prob\big(NAN_\mathrm{j} measures|0\rangle\big) \\ &= Prob\big(SM_\mathrm{i} measures|1\rangle\big)\big((Prob(NAN_\mathrm{j} measures \\ &\quad |0\rangle)|(Prob(SM_\mathrm{i} measures|1\rangle)\big) \\ &= \big(\langle e_1|e_1\rangle\big) * \frac{\langle e^0_{1,1}|e^0_{1,1}\rangle}{\langle e_1|e_1\rangle} \\ &= \langle e^0_{1,1}|e^0_{1,1}\rangle\end{aligned} \tag{9}$$

Similarly, we may estimate the following:

$$\begin{aligned}p_{0,0,0} &= \langle e^0_{0,0}|e^0_{0,0}\rangle, \quad p_{0,0,1} = \langle e^1_{0,0}|e^1_{0,0}\rangle \\ p_{1,0,1} &= \langle e^1_{0,2}|e^1_{0,2}\rangle, \quad p_{1,1,1} = \langle e^1_{1,3}|e^1_{1,3}\rangle \\ p_{0,1,0} &= \langle e^0_{1,1}|e^0_{1,1}\rangle, \quad p_{1,1,0} = \langle e^0_{1,3}|e^0_{1,3}\rangle \\ p_{0,1,1} &= \langle e^1_{1,1}|e^1_{1,1}\rangle, p_{1,0,0} = \langle e^0_{0,2}|e^0_{0,2}\rangle\end{aligned}$$

In accordance with [28, Lemma 1], when analyzing the von Neumann entropy of a specific system configuration, it's

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important to note that the density operator ρ operates on H, where each $\sigma_{\rm j}$ represents a Hermitian, positively semi-definite operator with a unit trace,

$$S(\rho) = H(p_1, p_2, p_3, \ldots, p_{\rm n}) + \sum_{j=1}^{n} p_{\rm j} S(\sigma_{\rm j}) \tag{10}$$

Now, we must use Eq. (10) and perform the following calculation to determine the key rate,

$$S(SM_{\rm i}|A) = S(SM_{\rm i}A) - S(A) \tag{11}$$

Due to the high dimensionality of H_A, a method similar to the one employed in [30] to demonstrate the security of the BB84 protocol was utilized. This is based on the strong subadditivity of von Neumann entropy, which implies that for any tripartite system $H_X \otimes H_Y \otimes H_Z$,

$$S(X|Y) \geq S(X|YZ) \tag{12}$$

Let us consider the four-dimensional space $\mathrm{H_C}$. With the introduction of this new system $\mathrm{H_C}$, we can obtain,

$$\begin{aligned} &S(SM_{\rm i}|A) - H(SM_{\rm i}|SM_{\rm i}) \geq S(SM_{\rm i}|AC) \\ &- H(SM_{\rm i}|NAN_{\rm i}) \end{aligned} \tag{13}$$

Let H_C be spanned by the basis states $|C,0\rangle$, $|C,1\rangle$, $|W,0\rangle$, $|W,1\rangle$. We utilize the state $|C,i\rangle\langle|C,i|$ to represent the scenario where the raw key bits of $SM_{\rm i}$ and $NAN_{\rm j}$ are in agreement (i.e., they are "correct") and the qubit sent from $NAN_{\rm j}$ was flipped i times in the Z basis. We are presently considering only Z basis states, as X basis states do not contribute to the raw key. Consider an instance where, $NAN_{\rm j}$ sends $|0\rangle$, $SM_{\rm i}$ measures $|1\rangle$, and $NAN_{\rm j}$ measures $|1\rangle$. This scenario corresponds to the event $|C,i\rangle\langle C,i|$. In a similar fashion, for the state $|W,i\rangle\langle W,i|$, where the raw key bits of $NAN_{\rm j}$ and $SM_{\rm i}$ do not match (are considered "wrong") [31]. Incorporating this new system yields the mixed state as follows:

$$\begin{aligned} &\rho^{\rm S}_{\rm MIXED} \\ &= \frac{1}{2}|0\rangle\langle 0|_{SM_{\rm j}} \otimes \Big(|C,0\rangle \otimes |e^0_{0,0}\rangle\langle e^0_{0,0}|_{\rm TA} + \\ &|W,1\rangle \otimes |e^1_{0,0}\rangle\langle e^1_{0,0}|_{\rm TA} + |C,1\rangle|e^0_{0,2}\rangle\langle e^0_{0,2}|_{\rm TA} + \\ &|W,2\rangle \otimes |e^1_{0,2}\rangle\langle e^1_{0,2}|_{\rm TA}\Big) + \frac{1}{2}|1\rangle\langle 1|_{SM_{\rm j}} \otimes \\ &\Big(|W,2\rangle \otimes |e^0_{1,1}\rangle\langle e^0_{1,1}|_{\rm TA} + |C,1\rangle \otimes |e^1_{1,1}\rangle\langle e^1_{1,1}|_{\rm TA} \\ &+ |W,1\rangle \otimes |e^0_{1,3}\rangle\langle e^0_{1,3}|_{\rm TA} + \\ &|C,2\rangle \otimes |e^1_{1,3}\rangle\langle e^1_{1,3}|_{\rm TA}\Big) \end{aligned} \tag{14}$$

By selecting an appropriate basis, we can represent $\rho^{\rm S}_{\rm MIXED}$ as a diagonal matrix, with the diagonal elements taking the form of $\frac{1}{2}\langle e^{\rm k}_{\rm i,j}|e^{\rm k}_{\rm i,j}\rangle$. Thus we can write it as,

$$\begin{aligned} &S(SM_{\rm i}AC) \\ &= S(\rho_{\rm MIXED}) = H(\frac{1}{2}\langle e^0_{0,0}|e^0_{0,0}\rangle, \ldots, \langle e^1_{1,3}|e^1_{1,3}\rangle) \\ &= H(\frac{1}{2}p_{0,0,0}, \ldots, \frac{1}{2}p_{1,1,1}) \end{aligned} \tag{15}$$

Now tracing out $SM_{\rm i}$ from $\rho_{\rm MIXED}$ to bound $S(AC)$ yields:

$$\begin{aligned} \rho_{\rm AC} &= |C,0\rangle\langle C,0| \otimes \frac{1}{2}\sigma_1 + |C,1\rangle\langle C,1| \otimes \frac{1}{2}\sigma_2 \\ &+ |W,1\rangle\langle W,1| \otimes \frac{1}{2}\sigma_3 + |W,2\rangle\langle W,2| \otimes \frac{1}{2}\sigma_4 \end{aligned} \tag{16}$$

where $\sigma_{\rm i}$ are the positive semi-definite operators:

$$\begin{aligned} \sigma_1 &= |e^0_{0,0}\rangle\langle e^0_{0,0}|_{\rm TA}, \quad \sigma_2 = |e^0_{0,2}\rangle\langle e^0_{0,2}|_{\rm TA} \\ \sigma_3 &= |e^1_{0,0}\rangle\langle e^1_{1,3}|_{\rm TA}, \quad \sigma_4 = |e^0_{0,2}\rangle\langle e^0_{1,1}|_{\rm TA} \end{aligned}$$

Also, let's assume that $tr\sigma_{\rm j} \geq 0$ for all j = 1, 2, 3, 4. We can denote $t_{\rm j}$ as $tr\sigma_{\rm j}$ and introduce $\tilde{\sigma}_{\rm j}$ as $\sigma_{\rm j}$ divided by $t_{\rm j}$. With these definitions, we can express ρ_{AC} as follows:

$$\begin{aligned} \rho_{\rm AC} &= |C,0\rangle\langle C,0| \otimes \frac{1}{2}t_1\tilde{\sigma}_1 + |C,1\rangle\langle C,1| \otimes \frac{1}{2}t_2\tilde{\sigma}_2 \\ &+ |W,1\rangle\langle W,1| \otimes \frac{1}{2}t_3\tilde{\sigma}_3 + |W,2\rangle\langle W,2| \otimes \frac{1}{2}t_4\tilde{\sigma}_4 \end{aligned} \tag{17}$$

where, each $\tilde{\sigma}_{\rm j}$ represents a positive semi-definite operator with unit trace. Furthermore, given that $tr\rho_{\rm AC} = 1$ (which implies $t_1 + t_2 + t_3 + t_4 = 1$), we can apply Eq. (10) to the state described above, resulting in:

$$S(AC) = S(\rho_{\rm AC}) = H(\frac{1}{2}t_1 + \cdots + \frac{1}{2}t_4) + \frac{1}{2}\sum_{j=1}^{4} t_{\rm j} S(\tilde{\sigma}_{\rm j}) \tag{18}$$

Here, we made the assumption that $t_{\rm j} \geq 0$. Also, if $t_{\rm j} = 0$, it implies that $\sigma_{\rm j} = 0$. Hence, we can apply the same arguments as explained earlier by omitting $\sigma_{\rm j}$ from the density matrix description. Besides, it is known that $t_1 = tr\sigma_1 = |e^0_{0,0}\rangle\langle|e^0_{0,0}|_{\rm TA} + |e^1_{1,3}\rangle\langle|e^1_{1,3}|_{\rm TA} = p_{0,0,0} + p_{1,1,1}$, and the same relationship holds for the other $\sigma_{\rm j}$. Noting that each $\sigma_{\rm j}$ corresponds to a two-dimensional system, we can employ the simple bound $S(\tilde{\sigma}_{\rm j}) \leq 1$ to illustrate:

$$\begin{aligned} S(AC) \leq\ & H(\frac{1}{2}(p_{0,0,0} + p_{1,1,1}), \frac{1}{2}(p_{1,0,0} + p_{0,1,1}), \\ & \frac{1}{2}(p_{0,0,1} + p_{1,1,0}), \frac{1}{2}(p_{1,0,1} + p_{0,1,0})) \\ & + \frac{1}{2}(p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0}) \\ & + \frac{1}{2}(p_{1,1,0} + p_{1,1,0})S(\tilde{\sigma}_{\rm j}) \end{aligned} \tag{19}$$

When the noise level in the SG communication channel is minimal, it is expected that most of the values for $p_{\rm i,j,k}$ are low, with the exceptions of $p_{0,0,0}$ and $p_{1,1,1}$ exhibiting high values. Therefore, the sole remaining objective is to determine an upper limit for $S(\tilde{\sigma}_1)$. Hence, it is necessary to determine the eigenvalues of σ_1, and $\tilde{\sigma}_1$. We can write, $|e^0_{0,0}\rangle = \sqrt{p_{0,0,0}}|e\rangle$ and $|e^1_{1,3}\rangle = \alpha|e\rangle + \beta|\gamma\rangle$ where, $\alpha,\beta \in C$, $|\alpha^2| + |\beta^2| = 1$ and $\langle e|e\rangle = 1, \langle e|\gamma\rangle = 0$,

$$\alpha^2 + \beta^2 = \langle e^1_{1,3}|e^1_{1,3}\rangle = p_{1,1,1} \tag{20}$$

$$\sqrt{p_{0,0,0}} = \langle e^1_{1,3}|e^1_{1,3}\rangle \tag{21}$$

$$\left|\alpha^2\right| = \frac{\left|\langle e^1_{1,3}|e^1_{1,3}\rangle^2\right|}{p_{0,0,0}} \tag{22}$$

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The eigenvalues of σ_1 are denoted as $|e\rangle$ and $|\gamma\rangle$. Additionally, let's designate the eigenvalues of $\tilde{\sigma}_1$ as $\tilde{\lambda_+}$ and $\tilde{\lambda_-}$. By employing Eq. (22) and principles of matrix algebra, we derive the following:

$$\tilde{\lambda_\pm} = \frac{1}{2} \pm \frac{\sqrt{(p_{0,0,0} - p_{1,1,1})^2 + 4\left|\langle e^1_{1,3}|e^1_{1,3}\rangle\right|^2}}{2(p_{0,0,0} + p_{1,1,1})} \quad (23)$$

and so: $S(\tilde{\sigma}_1) = -\tilde{\lambda_+}\log\tilde{\lambda_+} - \tilde{\lambda_-}\log\tilde{\lambda_-} = h(\tilde{\lambda_+})$, a function relying on the quantity $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2 \geq 0$. If $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2$ tends to zero, $\lambda_\pm$ will tend to $\frac{1}{2}$, causing $S(\tilde{\sigma}_1)$ to increase (reaching its maximum value of $\frac{1}{2}$). To establish a lower bound for the key rate, $r \geq S(\mathrm{SM}_i\ \mathrm{AC}) - S(EA) - H(\mathrm{SM}_i\,|\,\mathrm{NAN}_j)$, a lower bound for $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2$ (which, in turn, upper bounds S(EA)) required to be calculated. If, $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2 \geq \theta$, then $S(\tilde{\sigma}_1) = \mathrm{h}(\tilde{\lambda_+}) \leq \mathrm{h}(\tilde{\lambda})$, where,

$$\tilde{\lambda} = \frac{1}{2} + \frac{\sqrt{(p_{0,0,0} - p_{1,1,1})^2 + 4\theta}}{2(p_{0,0,0} + p_{1,1,1})} \quad (24)$$

Until now, our analysis has solely focused on noise within the Z basis. Now, let's shift our focus to the X basis noise and proceed to establish a lower bound for the value $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2$.

• *Using X-basis Noise:* Given the initial transmission of an X basis state, either $|+\rangle$ or $|-\rangle$, by NAN_j, SM_i opts to reflect, and NAN_j then proceeds to conduct measurements in the X basis, allowing NAN_j to evaluate the channel noise in this specific basis. During this operation, SM_i utilizes an identity operator. Therefore, if NAN_j sends the state $|-\rangle$ or $|+\rangle$, and it passes through A twice, it becomes $|a_1\rangle = V|a\rangle$, where $V = U_\mathrm{F}\,U_\mathrm{E}$. Following the equations and steps mentioned above for Z basis noise, we can derive an upper bound for $S(EA)$ by observing that $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2 = Re^2(\langle e^1_{1,3}|e^1_{1,3}\rangle)^2 + Im^2(\langle e^1_{1,3}|e^1_{1,3}\rangle)^2 \geq Re^2(\langle e^1_{1,3}|e^1_{1,3}\rangle)^2$. By considering $B = B^2$ and $B \geq 0$, it becomes clear from the previous discussion that $|\langle e^1_{1,3}|e^1_{1,3}\rangle|^2 \geq B$. Leveraging this insight, along with Eq. (23) and the preceding discussion, we can establish an upper bound on $S(\tilde{\sigma_1})$, consequently leading to an upper bound on the quantity $S(EA)$. Next, our focus is on calculating the lower bound for the final key rate r.

• *Final Key Rate Bound:* Now, we aim to calculate $H(SM_\mathrm{i}|NAN_\mathrm{j}) = H(SM_\mathrm{i}, NAN_\mathrm{j}) - H(NAN_\mathrm{j})$. Fortunately, this can be readily achieved with the Z basis error rate estimate provided by SM_i. Let $p_{\mathrm{NAN_j}}(0)$ represent the probability of NAN_j's raw key bit being zero. This value can be expressed as:

$$p_{\mathrm{NAN_j}}(0) = \frac{1}{2}(p_{0,0,0} + p_{0,1,0} + p_{1,1,0} + p_{1,0,0}) \quad (25)$$

and

$$H(NAN_\mathrm{j}) = H(p_{\mathrm{NAN_j}}(0), 1 - p_{\mathrm{NAN_j}}(0)) = h(p_{\mathrm{NAN_j}}(0)) \quad (26)$$

Consider $p(b, a)$ as the probability that the raw key bit of the SM_i is b, and the NAN_j's bit is a. Bringing all the components together, we can establish a lower bound for the key rate r as follows:

$$\begin{aligned} r \geq\ & H\left(\frac{1}{2}p_{0,0,0}, \ldots, \frac{1}{2}p_{1,1,1}\right) - H\left(\frac{1}{2}\big(p_{0,0,0} + p_{1,1,1}\big), \frac{1}{2}\right. \\ & \left.\big(p_{1,0,0} + p_{0,1,1}\big), \frac{1}{2}(p_{0,0,1} + p_{1,1,0}), \frac{1}{2}\big(p_{1,0,1} + p_{0,1,0}\big)\right) \\ & -\frac{1}{2}\big(p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0} + p_{1,1,0}\big) \\ & -\frac{1}{2}\big(p_{0,0,0} + p_{1,1,1}\big)h(\tilde{\lambda}) + h\big(p_{\mathrm{NAN_i}}(0)\big) \\ & -H(p(0,0), p(0,1), p(1,0), p(1,1)) \end{aligned} \quad (27)$$

where $\tilde{\lambda}$, from Eq. (23), is a function of SM_i computed in the previous section. This equation for key rate is a straightforward function that can be easily computed. It depends solely on parameters that can be estimated by both NAN_j and SM_i. The proof provided above validates the security against collective attacks. However, after the Q-Secure-P^2-SMA execution, SM_i and NAN_j can symmetrize their raw key by applying a randomly chosen permutation, which they publicly disclose. Importantly, ensuring security against collective attacks is enough to establish overall security against any arbitrary general attack. Hence, we have rigorously demonstrated the unconditional security proof for Q-Secure-P^2-SMA, ensuring its resilience against quantum threats.

Proposition 2: Q-Secure-P^2-SMA features Resistance to Entangle and Measure Attack.

Proof: In the entangle-measure attack, when the qubit travels from NAN_j to SM_i, the adversary A employs a unitary operation U_E to entangle it with an ancillary qubit $|e_\mathrm{i}\rangle$, resulting in the outcomes detailed in Eq. (3). The adversary then allows the traveling qubit to proceed on its path. After transmission, the adversary measures the ancillary qubit to gain information about SM_i's operations. Eq. (3) must satisfy specific conditions: a) $\langle e_0|e_0\rangle + \langle e_1|e_1\rangle = 1$; b) $\langle e_2|e_2\rangle + \langle e_3|e_3\rangle = 1$; c) $\langle e_0|e_1\rangle + \langle e_2|e_3\rangle = 0$; to pass eavesdropping detection successfully without introducing errors. These conditions ensure that the adversary cannot distinguish between $|e_{00}\rangle$ and $|e_{11}\rangle$. In Eq. (8), decoy qubits are inserted into the sequence at random, with the adversary unaware of their positions and initial states. Furthermore, SM_i reorders prior to returning decoy qubits to NAN_j, making it impossible for the adversary to ascertain the original order. As a result, the adversary is unable to extract valuable information from the ancillary qubit of the unitary operations, and the entangle-measure attack fails to provide any operational details. ■

Proposition 3: Q-Secure-P^2-SMA features Resistance to Trojan Horse Attack.

Proof: The Q-Secure-P^2-SMA functions as a bidirectional communication protocol, opening the possibility for potential attackers to employ the Trojan horse attack in an attempt to access confidential information. The Trojan horse attack typically falls into two categories: the delay photon Trojan horse attack and the invisible photon eavesdropping (IPE) attack. To safeguard against these attack strategies, Q-Secure-P^2-SMA assume that both SM_i and NAN_j incorporate photon number splitters and wavelength filters in front of their respective devices prior to engaging in communication [32].

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Proposition 4: Q-Secure-P^2-SMA features Mutual Authentication.

Proof: In the Q-Secure-P^2-SMA scheme, CC is responsible to authenticate SM_i and NAN_j. During Phase 1 of the Q-Secure-P^2-SMA, CC uses SQKD to pre-exchange the secret keys sequence $SharedKey_{SM_i}$ comprising $K(K_{2i-1}K_{2i} \in 00, 01, 10, 11, i = 1, 2, \ldots, n)$ with SM_i. CC generates a sequence of n Bell states, splitting them into two qubit sequences, S_H and S_T. CC also inserts $\frac{n}{2}$ decoy qubits (D) into S_T, forming the modified sequence S_{TD}, which is then shared with SM_i. Upon acknowledgment, SM_i discloses decoy qubit positions. Both parties, guided by $SharedKey_{SM_i}$, operate on these decoy qubits and compare results with initial states. If consistency persists, Z basis measurements occur on S_H and S_T qubits. Mutual authentication is then accomplished via bitwise XOR operations on measurement outcomes and $SharedKey_{SM_i}$, verifying both CC and SM_i.

Proposition 5: Q-Secure-P^2-SMA features Key Agrement.

Proof: As outlined in the phase 3 of Q-Secure-P^2-SMA scheme, SM_i upon receiving $AckPreAuthReq_{SM_i}$ from NAN_j, extracts $PSID_{SM_i}$ and proceeds to verify the authenticity of $PSID_{SM_i}$ by performing decryption with the $SharedKey_{SM_i}$. After successful $PSID_{SM_i}$ authenticity verification, SM_i and NAN_j uses SQKD to establish a $Sessionkey_{SM_i}$ corresponding to $PSID_{SM_i}$. ■

Proposition 6: Q-Secure-P^2-SMA features Resistance to Known Attacks.

Proof: Q-Secure-P^2-SMA resists various known attacks as follows:

- *Replay Attack:* Q-Secure-P^2-SMA scheme safeguards against replay attacks through various phases. During login and authentication, the transferred $PreAuthReq_{SM_i}$ is timestamped and authenticated with HMAC. In phase 4, the tuple $BillReq_{SM_i}$ comprising MSG and MCU is protected with HMAC and includes a timestamp T_{SM}, respectively. HMAC detects authentication errors, while T_{SM} leads to connection termination for attackers. To impersonate by using previous tokens, an attacker must tamper with timestamps. However, tampering with T_{SM} is challenging as it requires knowledge of $Sessionkey_{SM_i}$, rendering it difficult for attackers.
- *Impersonation Attacks:* To successfully impersonate the entity NAN_j, an attacker must circumvent the authentication process, which involves generating, preparing, and sending qubits. However, since the attacker lacks knowledge of the $SharedKey_{NAN_j}$, comprising a pre-shared sequence K, they can only generate decoy qubits at random from four possible states: $|0\rangle, |1\rangle, |+\rangle, and |-\rangle$. The probability of the attacker correctly guessing the basis for each decoy qubit is 1/4. If the attacker chooses the wrong basis, there is a 1/4 * 1/2 + 1/4 * 1/2 = 1/4 chance that the legitimate entity SM_i will obtain the correct measurement result. Consequently, the attacker has a 1/2 probability of evading detection and going unnoticed, i.e., if there are n/2 decoy qubits subject to the attacker's interference, the probability of detecting the attacker's attack is calculated as $P_1 = 1 - (\frac{1}{2})^{\frac{n}{2}}$. Likewise, when attempting to impersonate SM_i, the attacker cannot accurately manipulate the qubits within the sequence S_{TD} because they lack knowledge of the $SharedKey_{SM_i}$, comprising a pre-shared key sequence K. The attacker's likelihood of selecting the correct operation is 1/2. If the attacker selects the wrong operation, the legitimate entity NAN_j has a 1/4 chance of obtaining the correct measurement result. Consequently, the attacker can successfully evade eavesdropping detection with a probability of 3/4, i.e., if there are n/2 decoy qubits subject to the attacker's interference, the probability of detecting is calculated as $P_2 = 1 - (\frac{3}{4})^{\frac{n}{2}}$. It may be noted that when there is a substantial number (n) of decoy qubits in use, the detection probability $(P_1 and P_2)$ approaches 1 as n grows larger. Therefore, employing a sufficient quantity of decoy qubits makes it increasingly difficult for the attacker to avoid detection when attempting to impersonate either NAN_j or SM_i.
- *MITM Attacks:* In order to carry out a MITM attack, an adversary needs access to qubit sequences, specifically S_H, S_T, and S_{TD}. Afterwards, the adversary has to operate on the S_{TD} guided by $SharedKey_{SM_i}$ and $SharedKey_{NAN_j}$, corresponding to SM_i and NAN_j. Nonetheless, generating the precise secrets, $SharedKey_{SM_i}$ and $SharedKey_{NAN_j}$, corresponding to SM_i and NAN_j, necessitates knowledge of the CTRL bits and SHIFT bits. Also, the inherent randomness in choosing whether to measure or reflect the received qubits, as well as the random choice of sending the qubits in either the computational basis (Z) or the Hadamard basis (H), makes it virtually impossible to generate the exact secret keys. Consequently, the Q-Secure-P^2-SMA scheme effectively prevents and detects MITM attacks. ■

Proposition 7: Q-Secure-P^2-SMA features SM Anonymity.

Proof: The adversary's inability to access identification information $I_{Enc_{SM_i}}$ ensures anonymity for SM_i. During pseudo-identity generation, NAN_j creates a session key for $PSID_{SM_i}$ without storing data about $I_{Enc_{SM_i}}$. Furthermore, CC produces $PSID_{SM_i}$ by encrypting it with the secret key $SharedKey_{SM_i}$, known only to the trusted control center and SM_i, guaranteeing anonymity even from semi-trusted NAN_j. In the phase of smart meter data authentication and verification, SM_i sends the tuple $BillReq_{SM_i}$ to NAN_j for validation. This tuple includes MSG and MCU containing $PSID_{SM_i}$, which is salted and encrypted using a quantum secret exclusively shared between the CC and SM_i. As a result, it doesn't reveal any information about $I_{Enc_{SM_i}}$. ■

Proposition 8: Q-Secure-P^2-SMA features Forward Secrecy.

Proof: Upon receiving $ACKPreAuthReq_{SM_i}$ from NAN_j and validating $PSID_{SM_i}$, classical SM_i and quantum NAN_j engage in the generation of $Sessionkey_{SM_i}$ during the pseudo-identity creation phase for the smart meter. The randomness involved in deciding whether to measure or reflect received qubits, coupled with the random selection between the computational basis (Z) and the Hadamard basis (H) when sending qubits, renders it practically impossible to reproduce the exact

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session keys. Consequently, there exists no correlation between session keys from different sessions. Hence, it is unfeasible for an attacker to derive previous session keys from the current ones.

Proposition 9: Q-Secure-P^2-SMA features Message Unlinkability.

Proof: The adversary's inability to distinguish whether two tuples, $BillReq_{SM_i}$ and $BillReq^*_{SM_i}$, originate from the same SM_i or different ones results in SM_i message unlinkability. Let's assume that $BillReq_{SM_i} = (MSG,\ MD_{Current},\ MCU, PSID_{SM_i},\ T_{SM}) =$ HMAC$(Sessionkey_{SM_i},\ MD_{Current}, MCU, PSID_{SM_i}, T_{SM}))$, $MD_{Current}||PSID_{SM_i}||T_{SM})$, and $BillReq^*_{SM_i} = (MSG^*, MD_{Current},\ MCU^*,\ PSID_{SM_i},\ T_{SM})) =$ (HMAC$(Sessionkey_{SM_i},\ MCU^*)$, $MD^*_{Current}||\ PSID_{SM_i}||T_{SM}*)$. In simpler terms, $BillReq_{SM_i}$ and $BillReq^*_{SM_i}$ are indistinguishable even if they are transferred by the same SM_i. Both $BillReq_{SM_i}$ and $BillReq^*_{SM_i}$ are entirely unpredictable and random due to the fact that the HMAC function acts as a hash function that conforms to the random oracle property, meaning that even a single-bit change in its input results in uniformly random output. Also, in the Q-Secure-P^2-SMA, $BillReq_{SM_i}$ undergoes significant changes in each iteration. Additionally, $(MD_{Current}||\ PSID_{SM_i}||T_{SM})$ and $(MD_{Current}||PSID_{SM_i}||T^*_{SM})$ remain unrelated due to differences in timestamps. Typically, the metering data $(MD_{Current})$ for any SM_i changes every month. Likewise, MSG = HMAC$(Sessionkey_{SM_i}$, MCU) and $MSG^* =$ HMAC$(Sessionkey_{SM_i},\ MCU^*)$ remain unrelated due to variations in timestamps and/or metering data $(MD_{Current})$. Consequently, the adversary cannot establish a link between $BillReq_{SM_i}$ and $BillReq^*_{SM_i}$, ensuring the unlinkability of messages and, in turn, preserving the untraceability of SM_i. Thus, the Q-Secure-P^2-SMA prioritizes privacy preservation.

Proposition 10: Q-Secure-P^2-SMA features SM Identity Privacy.

Proof: During the phase 3 of Q-Secure-P^2-SMA, the control center produces $PSID_{SM_i}$ by encrypting the received identity $I_{Enc_{SM_i}}$ (derived from encoding I_{SM_i} and the I_{SM_i}'s owner biological characteristics). This encryption is performed using a quantum secret shared exclusively between CC and SM_i during the registration process. Subsequently, SM_i utilizes $PSID_{SM_i}$ in phase 4 of the Q-Secure-P^2-SMA for metering data reporting to NAN_j, thereby preserving I_{SM_i}. Furthermore, during login and authentication, SM_i forwards $I_{Enc_{SM_i}}$ for authentication purposes, maintaining the confidentiality of I_{SM_i}. While the semi-trusted NAN_j might be inclined to inquire about I_{SM_i} since SM_i receives $PSID_{SM_i}$ through NAN_j, it is infeasible for NAN_j to extract the real identity of SM_i. This is due to the fact that I_{SM_i} is encoded with the biological characteristics of the owner of I_{SM_i} and encrypted using a quantum secret, $SharedKey_{SM_i}$, which is exclusively shared between CC and SM_i. Additionally, the disclosed $PSID_{SM_i}$ does not contain any information related to SM_i as it incorporates salt, rendering it useless even to an inquisitive NAN gateway. ■

Proposition 11: Q-Secure-P^2-SMA features Unconditional Security.

TABLE IV
SECURITY FEATURE COMPARISON

SF	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
[4]	Yes	No	No	Yes	No	No	Yes	Yes	No	Yes	No
[5]	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No	No
[6]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No
[8]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
[10]	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
[11]	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	No
[33]	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No
[34]	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No
[18]	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	No
[22]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ours	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

SF: **Security Features** F1: **Mutual Authentication** F2: **Session Key Security**
F3: **Resists MITM Attack** F4: **Replay Protection** F5: **Resists Impersonation Attack**
F6: **Message Integrity** F7: **Anonymity** F8: **Forward Secrecy**
F9: **Unconditional Security** F10: **Privacy** F11: **Untraceability**

Proof: The Q-Secure-P^2-SMA scheme harnesses the fundamental principles of quantum mechanics and employs Bell states and measurement operations while exploiting the entangled properties of Bell states for the secret key generation; $SharedKey_{SM_i}$ and $SharedKey_{NAN_j}$. Importantly, the security of Q-Secure-P^2-SMA scheme is independent of computational hard problems like discrete logarithm or integer factorization problems, as well as quantum hard problems such as the shortest vector problem or closest vector problem. Consequently, it offers unconditional security. ■

B. Security Features Comparison

We examined related articles, and an assessment of security features is presented in Table IV.

VI. Performance Analysis & Simulation

This section provides the performance analysis of Q-Secure-P^2-SMA with reference to existing works [4], [5], [8], [10], [11], [18], [22]. The analysis encompasses three essential aspects: energy overhead, communication overhead and computation overhead. The availability of real smart grid implementation environments with quantum communication capabilities is limited. However, simulators serve as the most effective means to replicate a real-world environment. Qiskit [35] is an open-source software development kit designed for working with quantum computers at various levels, including circuit, pulse, and algorithm. It provides a suite of tools enabling the creation, manipulation, and execution of quantum programs either on the IBM Quantum Platform's prototype quantum devices or on local computer simulators. One such simulator, the Qasm Simulator, is part of the Qiskit-Aer package and emulates quantum circuit execution on real noisy devices. We used Qiskit version 0.43.1 and Qiskit-Aer version 0.12.0 for Q-Secure-P^2-SMA simulations. These simulations were carried out on a system with Windows 11 OS using Python (version 3.9.13) as the programming language, on the desktop with hardware configuration; 11th Gen Intel Core i5-1155G7 @ 2.50GHz processor with 8GB of RAM. Fig. 2 represents the circuit diagram for shared key: 1011; corresponding to phase 2 of Q-Secure-P^2-SMA. Besides, the specific details of computation overhead, communication overhead and energy utilization of the proposed Q-Secure-P^2-SMA are as follows:

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TABLE V
COMPUTATION OVERHEAD

Scheme	SM	NAN Gateway / CC	Total Computation Cost
[4]	$4T_m + 1T_{add} + 1T_{ex} + 5T_h$	$3T_m + 2T_{bp} + 1T_{add} + 1T_{ex} + 5T_h$	$7T_m + 2T_{add} + 2t_{bp} + 2T_{ex} + 10T_h$
[5]	$3T_m + 3T_{add} + 1T_{ex} + 6T_h$	$2T_m + 2T_{bp} + 3T_{add} + 1T_{ex} + 6T_h$	$5T_m + 6T_{add} + 2T_{bp} + 2T_{ex} + 12T_h$
[8]	$4T_m + 2T_{ed} + 2T_{mac} + 4T_h$	$3T_m + 2T_{ed} + 2T_{mac} + 5T_H$	$6T_m + 4T_{ed} + 4T_{mac} + 9T_h$
[10]	$2T_m + 4T_h$	$2T_m + 4T_h$	$4T_m + 8T_h$
[11]	$4T_m + T_{add} + 4T_h$	$4T_m + T_{add} + 4T_h$	$8T_m + 2T_{add} + 8T_h$
[18]	$2T_m + 2T_{add} + 3T_h + 2T_E + T_D$	$2T_m + 3T_{add} + 5T_h + 2T_E + T_D$	$4T_m + 5T_{add} + 8T_h + 4T_E + 2T_D$
[22]	$2T_{Hmac} + T_D$	T_E	$2T_{Hmac} + T_E + T_D$
Q-Secure-P^2-SMA	$2T_{Hmac} + T_D$	T_E	$2T_{Hmac} + T_E + T_D$

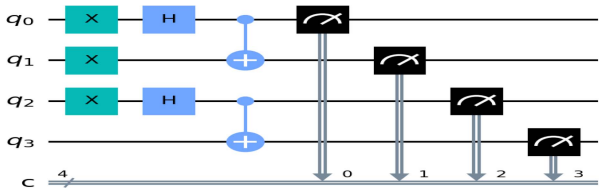

Fig. 2. Circuit Diagram: Phase 2 of Q-Secure-P^2-SMA.

TABLE VI
COMMUNICATION AND ENERGY OVERHEAD

Scheme	Communication Overhead (bits)		Energy Consumption (μJ)
	Sent Overhead	Received Overhead	
[4]	2240	1184	2572
[5]	1248	672	1443
[8]	544	544	832
[10]	157	352	398.16
[11]	672	352	768.96
[18]	3868	3070	5271.6
[22]	424	170	442.98
Q-Secure-P^2-SMA	632	160	584.64

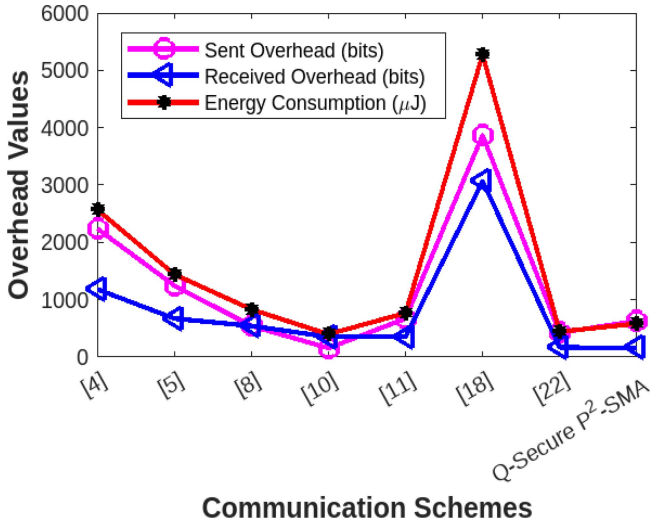

Fig. 3. Communication Overhead Analysis.

A. Communication Overhead

To compute the Q-Secure-P^2-SMA's communication overhead, we considered the message sizes and the quantity of messages exchanged between SM and NAN_j, as well as between NAN_j and CC. The assumed bit lengths of various variables in the transmitted SG messages are as follows: Hash values = 16B, identity = 1B, metering data = 1B, random number = 8B, MAC = 4B, and time stamp = 4B [8]. We categorize the transmitted messages into two groups: the sent overhead (comprising the message overhead sent from the SM) and the received overhead (comprising the message overhead received by the SM). In Q-Secure-P^2-SMA's phase 2, SM_i transmits $PreAuthReq_{SM_i}$ to confirm its identity, resulting in a communication overhead of 21 bytes. In the subsequent phase, which involves the generation of pseudo-identities, SM_i receives $AckPreAuthReq_{SM_i}$ from CC via NAN_j, incurring a communication overhead of 20 bytes. Finally, in phase 4 of Q-Secure-P^2-SMA, SM_i dispatches the tuple $BillReq_{SM_i}$ to NAN_j for metering data reporting, introducing a communication overhead of 58 bytes. The specifics of the sent overhead and received overhead costs for Q-Secure-P^2-SMA are provided in Table VI and illustrated in Fig. 3. Clearly, Q-Secure-P^2-SMA features lower communication overhead compared to state-of-art works [4], [5], [8], [11], [18]. Although, Q-Secure-P^2-SMA has slightly higher overhead compared to [10], [22], it facilitates quantum-secure metering data communication even with classical SM_i, unlike [10], [22].

B. Computation Overhead

Q-Secure-P^2-SMA considers the execution time associated with ECC point multiplication (T_m), encryption (T_E), and decryption (T_D), one-way hash (T_h), and MAC operation (T_{MAC}) as 2900ms, 3.8ms, 41.1ms, 39ms, and 8.6 ms, respectively [10]. In the second phase of the Q-Secure-P^2-SMA, a computational cost T_{HMAC} is incurred as a result of utilizing HMAC within the transmitted tuple $PreAuthReq_{SM_i}$ associated with smart meter login and authentication. Afterwards, in the third phase of Q-Secure-P^2-SMA, two critical operations take place: the generation of pseudo-identities for SM_i and the subsequent transfer of this pseudo-identity within $AckPreAuthReq_{SM_i}$ back to SM_i. These operations introduce computational expenses equivalent to T_E during the generation of the SM_i's pseudo-identity and T_D for the verification of the authenticity of the received $PSID_{SM_i}$, achieved by decrypting it from $AckPreAuthReq_{SM_i}$ by SM_i. In the phase 4, which revolves around the reporting of metering data, a computational cost emerges equivalent to the execution time of T_{HMAC}. This cost arises as the SM_i calculates the message to be included in the $BillReq_{SM_i}$ tuple. Accordingly, Q-Secure-P^2-SMA's total computation cost amounts to $2T_{HMAC} + T_E + T_D$. Table V presents a relative comparison of computational costs with existing schemes. While Q-Secure-P^2-SMA carries a equivalent computational overhead as [22], it distinguishes itself by its practicality. Unlike [22], Q-Secure-P^2-SMA doesn't require SM_i to possess full quantum capabilities; instead, it

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facilitates quantum-secure metering data communication even with classical SM_i.

C. Energy Overhead

The overall energy consumption of the Q-Secure-P^2-SMA is computed by considering the energy consumed by SM_i during both communication and computation overhead. Specifically, energy consumption is determined by the product of voltage (*vol*), current (*cur*), and execution time (*T*) associated with various cryptographic primitives utilized in Q-Secure-P^2-SMA. We consider a TelosB *SM* with $vol = 3v \& cur = 1.8\mu A$ for computing the energy overhead [10]. Additionally, the energy consumption for receiving and sending 1 bit of data by *SM* is estimated to be $0.81\mu J$ and $0.72\mu J$ respectively [10]. Q-Secure-P^2-SMA consumes approximately $411.84\mu J$ of energy due to communication overhead. Table VI presents a comparison of the energy overhead with existing work.

VII. Conclusion

This work introduces Q-Secure-P^2-SMA, a novel privacy-preserving mutual authentication scheme designed for facilitating quantum-secure communication within the smart metering infrastructure. The scheme employs semi-quantum key distribution alongside classical identity authentication to establish mutual authentication and communication among the various smart grid entities. After successful authentication of SM_i using semi-quantum key distribution, the control center generates and conveys a pseudo-identity corresponding to SM_i through the NAN gateway. Subsequently, SM_i and the NAN gateway generate a session key for the received pseudo-identity, using the semi-quantum key distribution protocol. The session keys are utilized by SM_i to ensure secure and privacy-preserving metering data reporting. Q-Secure-P^2-SMA's security analysis demonstrates its robustness against a range of potential attacks, including replay attacks, measure and resend attacks and impersonation attacks. Additionally, it provides SM_i identity privacy, SM_i anonymity, and SM's message unlinkability. Furthermore, a comparative analysis of overhead, focusing on communication and computation, reveals that Q-Secure-P^2-SMA is lightweight and exhibits reasonable energy consumption levels, all while delivering unconditional security. Q-Secure-P^2-SMA utilizes semi-quantum key distribution, enabling quantum secure communication between classical and quantum entities, making it well-suited for existing real-world smart grid scenarios, offering unconditional security without the need to update existing smart meters for quantum security. In future work, the capabilities of Q-Secure-P^2-SMA could be extended to support decentralized authentication of SG entities by leveraging blockchain technology.

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